

API TESTING MASTERCLASS

API Testing Cheat Sheet

The essentials of HTTP, requests and assertions on one sheet.

This is your one-page working reference for API testing over HTTP. Keep it beside you when writing requests in Postman or automating them in code: it covers the anatomy of a request, the methods and headers you use daily, how to read a response, and the assertions that turn a manual check into a repeatable test.

Anatomy of a Request

Every HTTP request is made of four parts. Get these right and the rest follows.

- Method — the verb: what you want to do (GET, POST, PUT, PATCH, DELETE).
- URL — the endpoint plus any path and query parameters, e.g. <https://api.example.com/users/1?expand=orders>
- Headers — metadata such as Authorization, Content-Type and Accept.
- Body — the payload you send (usually JSON), used by POST, PUT and PATCH.

HTTP Methods

Method	Purpose	Idempotent	Body
GET	Retrieve a resource; must not change state	Yes	No
POST	Create a resource or trigger an action	No	Yes
PUT	Replace a resource in full	Yes	Yes
PATCH	Partially update a resource	No	Yes
DELETE	Remove a resource	Yes	Optional
HEAD	Like GET but headers only, no body	Yes	No

Idempotent means repeating the same call has the same effect as making it once — a key property to test, especially for retries.

Key Headers

Header	Direction	What it does
Authorization	Request	Credentials, e.g. "Bearer <token>" or Basic auth

Header	Direction	What it does
Content-Type	Both	Format of the body being sent, e.g. application/json
Accept	Request	Format the client wants back, e.g. application/json
Content-Length	Both	Size of the body in bytes
Cache-Control	Both	Caching rules, e.g. no-cache
Location	Response	URL of a newly created or moved resource

The Response

A response mirrors the request: a status line, headers and (usually) a body.

- Status code — the outcome as a number, e.g. 200 OK or 404 Not Found (see the Status Codes reference).
- Headers — metadata about the response, e.g. Content-Type, Location, caching.
- Body — the returned data, typically JSON. Always confirm the body, not just the status.

Postman Assertions

Assertions live in Postman's Scripts (Tests) tab and run after the response arrives. These are the ones you reach for most.

Goal	Snippet
Assert status code	<code>pm.response.to.have.status(200);</code>
Response time under 500 ms	<code>pm.expect(pm.response.responseTime).to.be.below(500);</code>
Parse the JSON body	<code>const json = pm.response.json();</code>
Field equals a value	<code>pm.expect(json.id).to.eql(1);</code>
Field exists	<code>pm.expect(json).to.have.property('email');</code>
Array has a length	<code>pm.expect(json.items).to.have.lengthOf(3);</code>
Header present	<code>pm.response.to.have.header('Content-Type');</code>

Wrap related checks in `pm.test('a clear name', () => { ... });` so the results read like a report.

INSIDE STLC PRO TIP

Never assert on status alone. A 200 with the wrong body, a missing field or a five-second response time is still a failing test. Check the status, then the shape and values of the body, then performance where it matters.